

Analysis of iPSC Gene Expression (will be changed)

Final Project

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Abstract

Introduction

The discovery of human induced pluripotent stem cells (iPSCs) by Takahashi and Yamanaka in 2007 (Takahashi et al., 2007) has revolutionized the regenerative medicine field. iPSCs are created by reprogramming somatic cells into an embryonic like state using the Yamanaka transcription factors: SOX2, OCT4, KLF4, and c-Myc). These cells have vast capabilities due to their ability to model human development and diseases *in vitro*, and their potential for patient specific (autologous) and “off the shelf” (allogeneic) cell therapies (Cerneckis et al., 2024).

With that being said, the characterization of iPSC lines is essential in any experimental or manufacturing process to ensure consistency in culturing. This includes analyzing the expression of pluripotency (stem cell) markers, differentiation markers, and potential chromosomal abnormalities, to assess the population of the culture. This is commonly done using techniques such as reverse transcriptase or real time quantitative PCR (RT-qPCR) to analyze mRNA expression (Artyukhov et al., 2017), or using an antibody based experiment to analyze protein expression levels, such as Flow Cytometry or immunocytochemistry. When conducting these experiments, specifically RT-qPCR, reference genes, also known as housekeeping genes, are used to normalize the expression of other genes in a sample (Artyukhov et al., 2017). This allows the expression of genes to be compared across different samples, since it is presumed that the expression of the housekeeping gene is consistent across all of the samples. In stem cell samples, however, it has been found that the commonly used housekeeping genes, such as GAPDH, appear to fluctuate in iPSC samples (Panina et al., 2020), making it difficult to achieve accurate normalization data.

RNA Seq is a technique that sequences the entirety of RNA molecules in a given sample, rather than using gene specific primers or probes as in RT-qPCR and other techniques. This allows the entire transcriptome to be analyzed, providing a complete picture of the expression of genes in a sample (Wang et al., 2009). Additionally, gene expression is calculated by dividing the raw counts of a gene by the total counts in a sample, eliminating the need for a housekeeping gene. This allows for an unbiased and quantitative analysis of millions of genes in sample (Wang et al., 2009). Despite these advantages, RNA-Seq is far more time consuming and expensive than RT-qPCR and other experiments, making it non practical in a manufacturing setting. Therefore, utilizing RNA-Seq to analyze iPSC samples could provide insight into housekeeping gene expression, allowing researchers to select ideal reference genes for their analysis.

A similar study, conducted by Carcamo-Orive *et al.*, investigated the variability of gene expression across 317 iPSC lines generated from 101 individuals (Carcamo-Orive et al., 2017) and created a publically available data set of their raw RNA Seq data. They found that genetic variability in the iPSC lines was influenced by a donor specific gene expression pattern, specifically influenced by donor age, sex, ancestry, and reprogramming variabilities. iPSCs can also be reprogrammed from a variety of somatic cells, such as blood, skin fibroblasts, hair keratinocytes, urine cells, mesencymal stem cells, and bone marrow (Poetsch et al., 2022). While some somatic cells are easier to obtain than others, such as blood cells and fibroblasts, iPSCs from these lines are associated with higher genomic mutations due to the somatic cells' high turnover rate. Additionally, iPSCs appear to have an epigenetic memory of their tissue of origin, which has been found to influence their differentiation (Poetsch et al., 2022). All of these factors, and many others, could be contributing to the variations in gene expression in iPSC samples.

This publicly available dataset from Carcamo-Orive *et al.* was used to analyze the expression of RNA across iPSC samples, with a focus on the expression of pluripotency markers, differentiation markers, and housekeeping genes.

Methods

Data Cleaning

RNA-Seq counts per million data (named: `cpm_data`), and sample metadata (named: `raw_counts`) were acquired from Stemformatics using the Carcamo-Orive (2017) dataset (Carcamo-Orive et al., 2017). Genes in this study were referred to by their Ensembl ID, so gene information, such as name and RNA type, was integrated from the Ensembl database. Using R (R Core Team, 2025), the raw count per million data was converted to a log base 2 scale with a 0.5 offset to prevent samples with a count of 0 from becoming skewed. Values for age, from the metadata file, were grouped by decade, and an additional column was added for an age + sex category (i.e., 40s female). Information from the sample description was then

extracted to create columns for race and insulin resistance. The metadata and log counts per million files were then merged into one data frame for further analysis.

Category Tables

The number of samples for each group within the age-sex and race categories were summarized and displayed as two tables using the knitr (Xie, 2025) and kableExtra (Zhu, 2024) packages.

Pie Charts by RNA Category

Genes detected in the RNA Seq counts per million data were grouped into the following categories: Protein coding, long non-coding RNA (lncRNA), Small RNAs, Pseudogenes, and Other. A count of the number of genes in each category was then performed by using the dplyr package (Wickham et al., 2023) from tidyverse (Wickham, 2023). The total expression of each category was also calculated using the raw count per million data. The scales (Wickham, Pedersen, et al., 2025) package from ggplot2 (Wickham, Chang, et al., 2025) was used to create two pie charts, one of the gene counts and one of the gene expression by RNA category, plotted together using the patchwork package (Pedersen, 2025).

Principal Component Analysis (PCA) Plots

Log expression values were first cleaned by removing entries without gene information and duplicate gene names. Gene variances were calculated across the samples, and genes with no variance were removed. To visualize the major sources of variation in the expression data, a principal component analysis (PCA) was performed using the FactoMineR package (Husson et al., 2025), which identifies the major axes of variation in the dataset. To assess if sample characteristics (sex, age decade, combined age sex groups, and race) influenced the PCA's structure, a Welch's one-way ANOVA, which does not assume equal variance, was performed separately for PCA1 and PCA2. Significant differences were found for the age sex groups and race categories, so two PCA biplots were generated, each overlaying confidence ellipses corresponding to either age sex groups or race. On both of the biplots, the top twenty genes with the highest variance across all samples were labeled and displayed as vectors.

Pluripotency, Differentiation, and Housekeeping Gene Expression

The log expression of genes, categorized as pluripotency markers, differentiation markers, and housekeeping genes, was compared across sample groups (age-sex and race) using heatmaps, linear mixed-effects models, and Welch's ANOVA with post hoc Games-Howell tests. Heatmaps of log gene expression were generated using the pheatmap package (Kolde, 2025) to visualize overall expression patterns within each gene category. Linear mixed-effect models were then

fit using lme4 (Bates et al., 2025) and lmerTest (Kuznetsova et al., 2020) to evaluate differences in gene expression across age-sex and race groups, while accounting for gene-specific baseline expression levels by using random intercepts for genes. Percent differences in overall gene expression were calculated by exponentiating the estimated log scale coefficient back to fold change, subtracting one, and multiplying by 100. QQ plots and histograms were generated, but not shown here, for each model to view residuals and assess model assumptions. To identify which genes displayed statistically significant differences in expression between sample categories, Welch’s one-way ANOVA, assuming unequal variance, was performed for each gene. P values were adjusted using the Benjamini-Hochberg method due to the high number of tests that were performed. For genes with significant ANOVA results, Games-Howell post hoc tests, using rstatix (**R-statix?**), were performed and p-values were Benjamini-Hochberg corrected. Significant Games Howell comparisons (adjusted $p < 0.05$) were annotated on bar plots displaying log expression for each gene across sample groups.

Results

Table 1: Sample counts by Age-Sex

Age-Sex Group	Sample Count
40s female	34
40s male	8
50s female	63
50s male	41
60s female	77
60s male	57
70s female	17
70s male	11

Table 2: Sample counts by Race

Race	Sample Count
African American	20
Asian	33
Caucasian	212
Hispanic	14
South Asian	13
NA	16

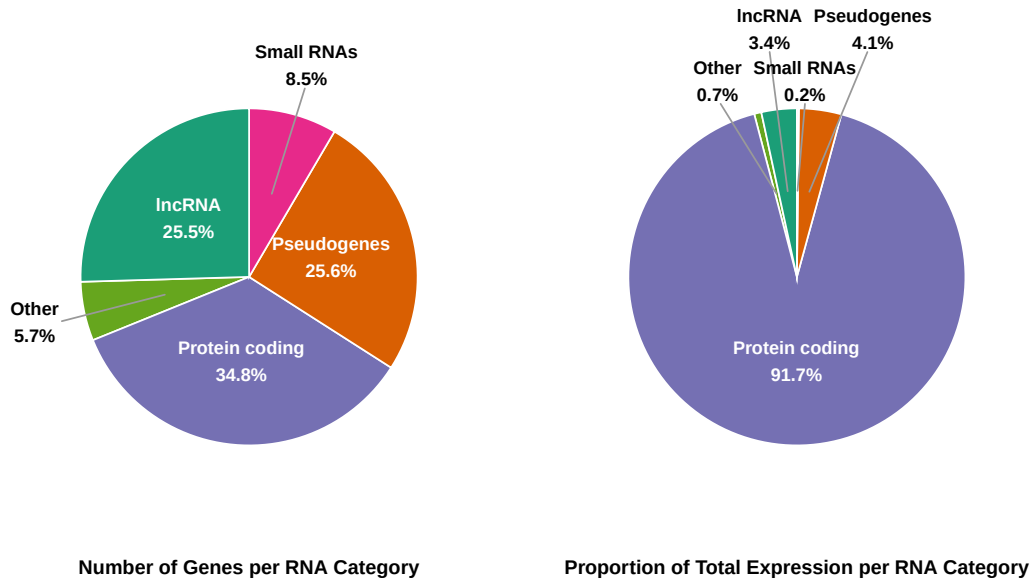


Figure 1: Detected RNA Types in iPSC Samples Analyzed using RNA Seq. The number of genes detected in each RNA category from RNA-seq data were counted and plotted as a pie chart (left). Small categories ($< 2\%$), such as rRNA, tRNA, and immunoglobulin genes, were combined into an ‘Other’ category. The expression of genes, using raw count per million data, was averaged by type of RNA and plotted as a pie chart (right).

PCA – Biplot

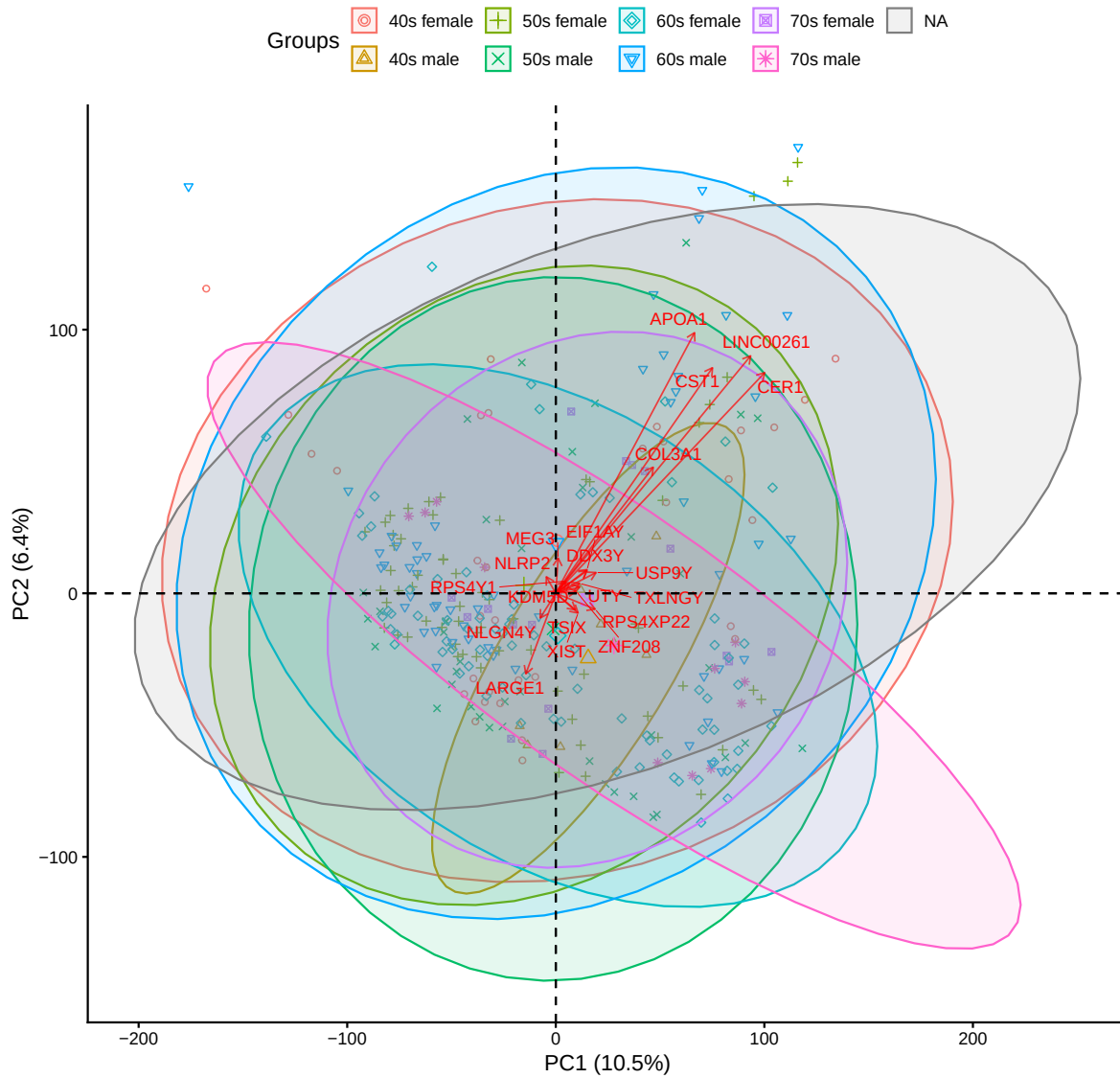


Figure 2: Principal Component Analysis (PCA) grouped by Donor's Age and Sex. PCA of log2 gene expression across all samples. PCA1 and PCA2 explain 10.5% and 6.4% of the variance, respectively. Samples are colored by age-sex groups and ellipses represent 95% confidence intervals for each group. Twenty genes contributing most to the variance are labeled as vectors.

PCA – Biplot

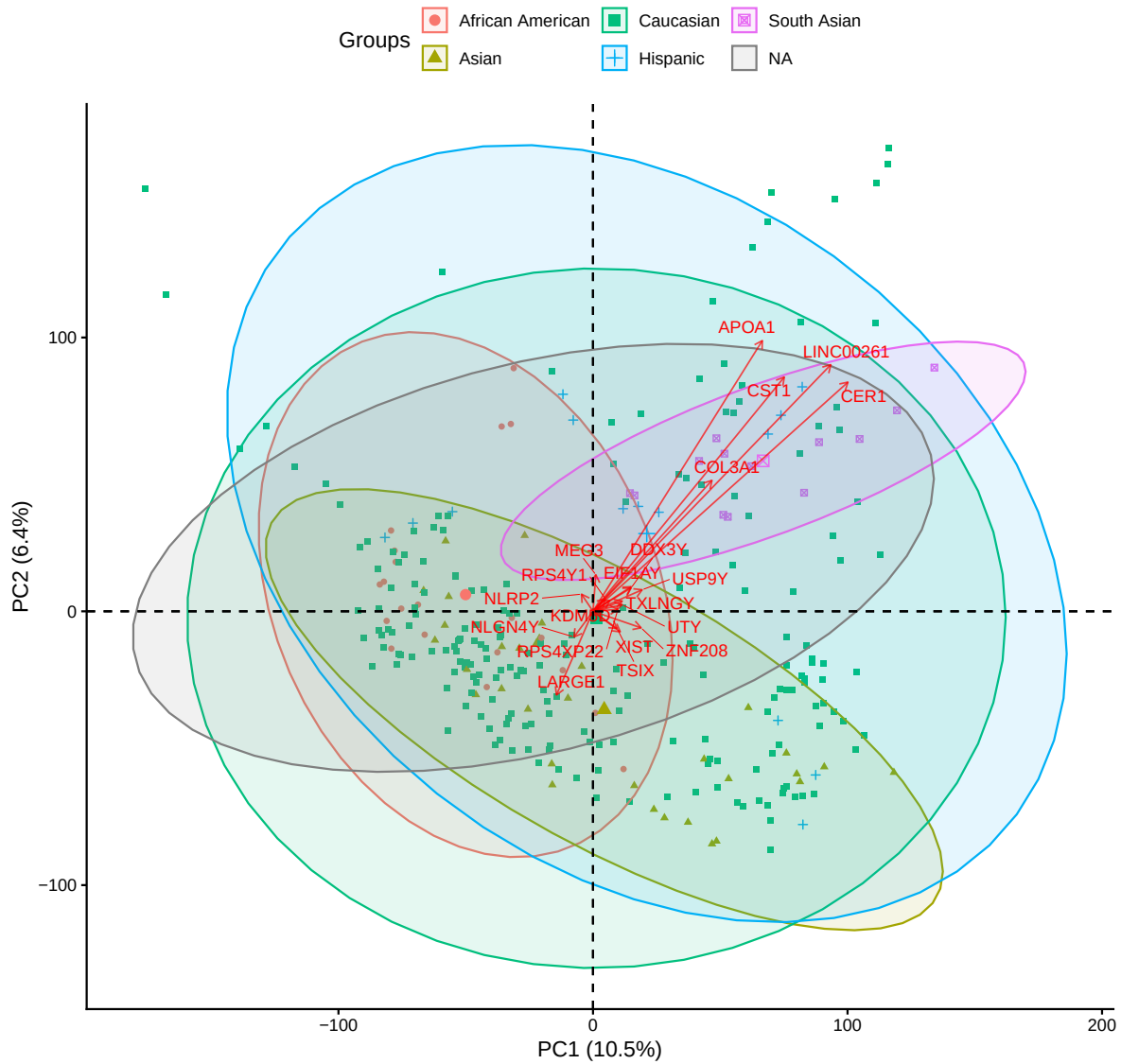


Figure 3: Principal Component Analysis (PCA) grouped by Donor's Race. PCA of log2 gene expression across all samples. PCA1 and PCA2 explain 10.5% and 6.4% of the variance, respectively. Samples are colored by race groups and ellipses represent 95% confidence intervals for each group. Twenty genes contributing most to the variance are labeled as vectors.

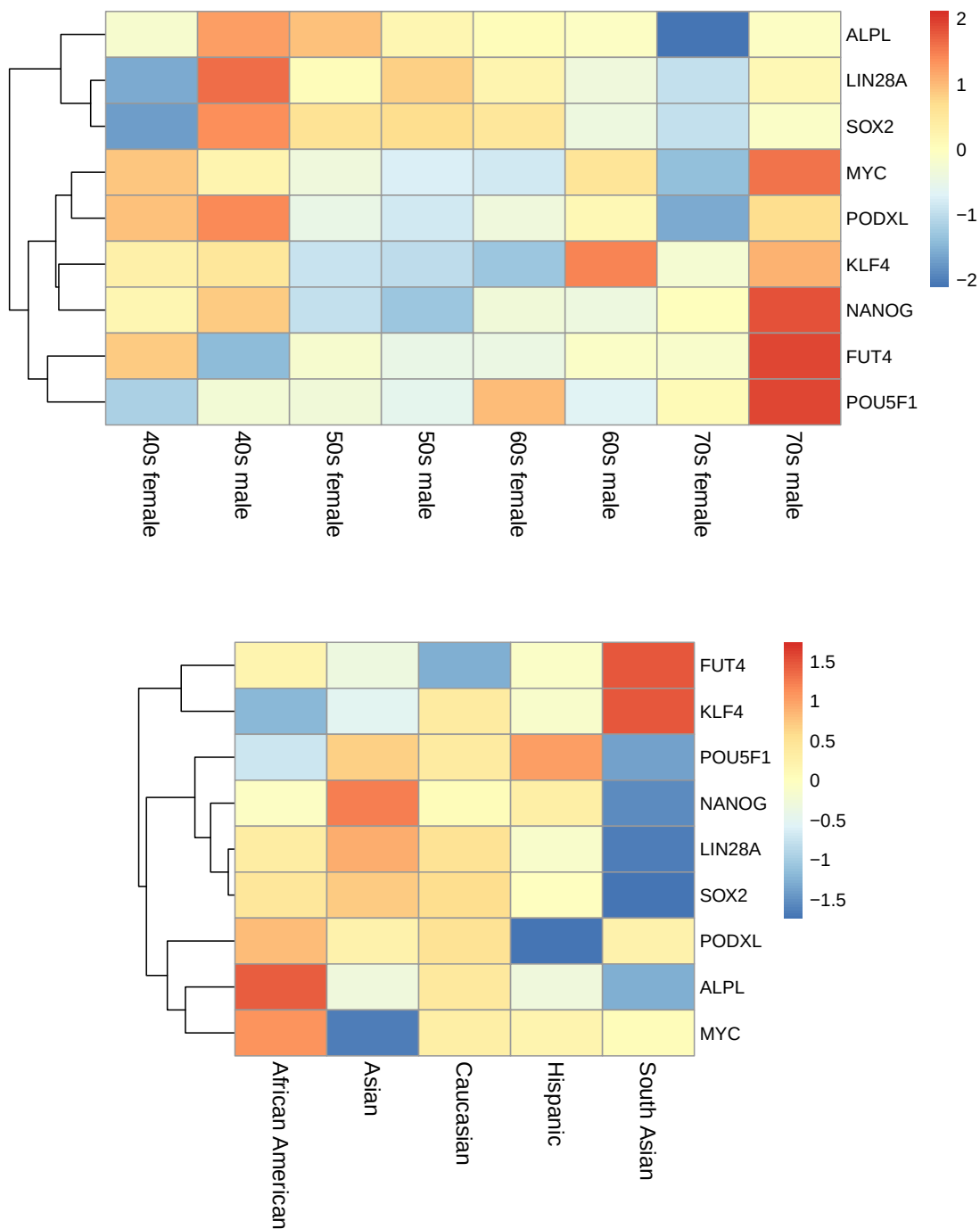


Figure 4: Heatmap of Pluripotency Gene Expression. Rows correspond to genes and columns correspond to samples grouped by age-sex (top) and race (bottom). Colors indicate relative expression levels based on log2 count per million expression. Hierarchical clustering was applied to the genes to highlight patterns of expression.

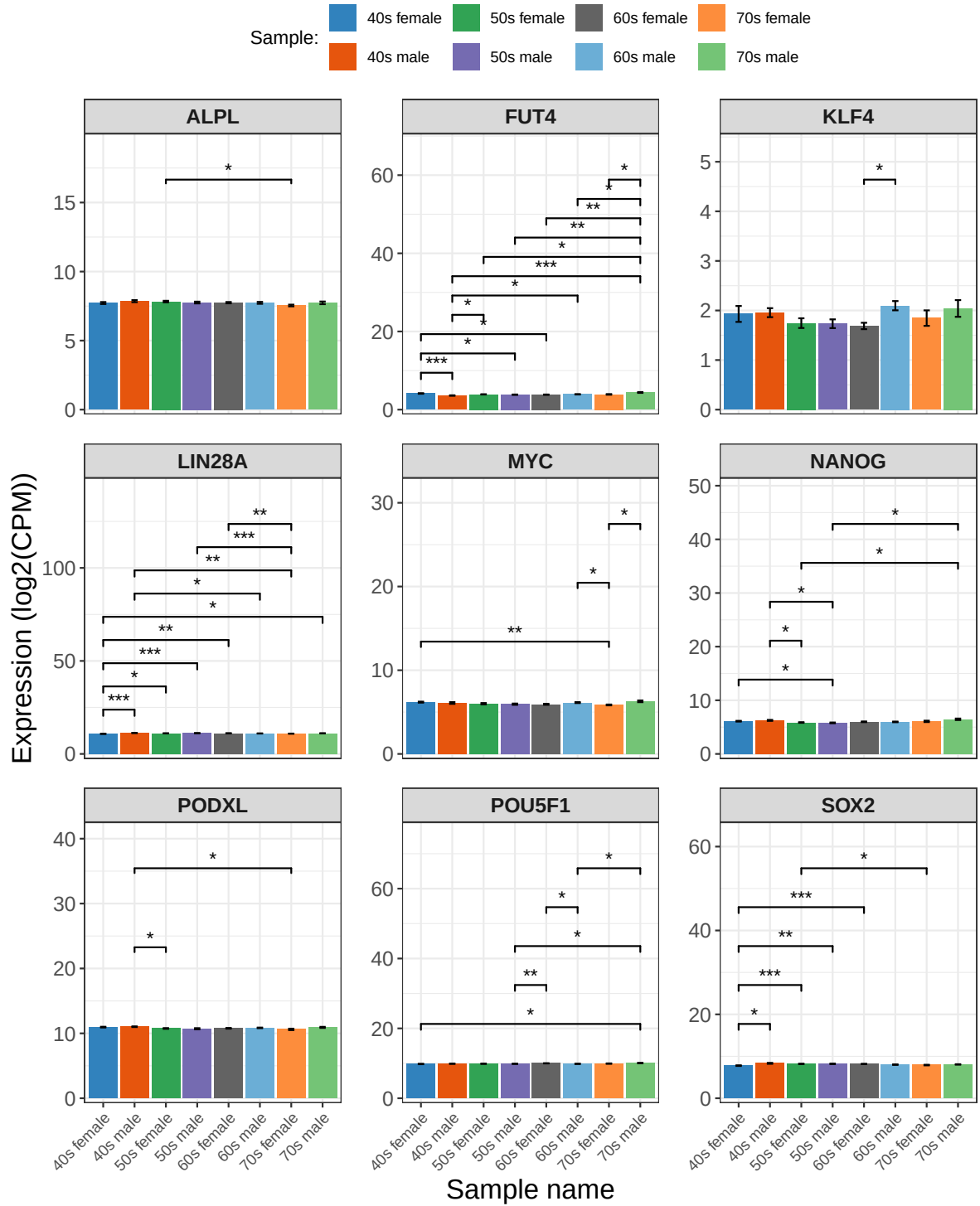


Figure 5: Expression of Pluripotency Genes by Age-Sex. Bar plots of log2 count per million gene expression by pluripotency gene. Bars represent mean expression $\pm$ standard error. Pairwise statistical significance was determined using Games-Howell post hoc tests following Welch's ANOVA, and significance was indicated with asterisks: $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***).

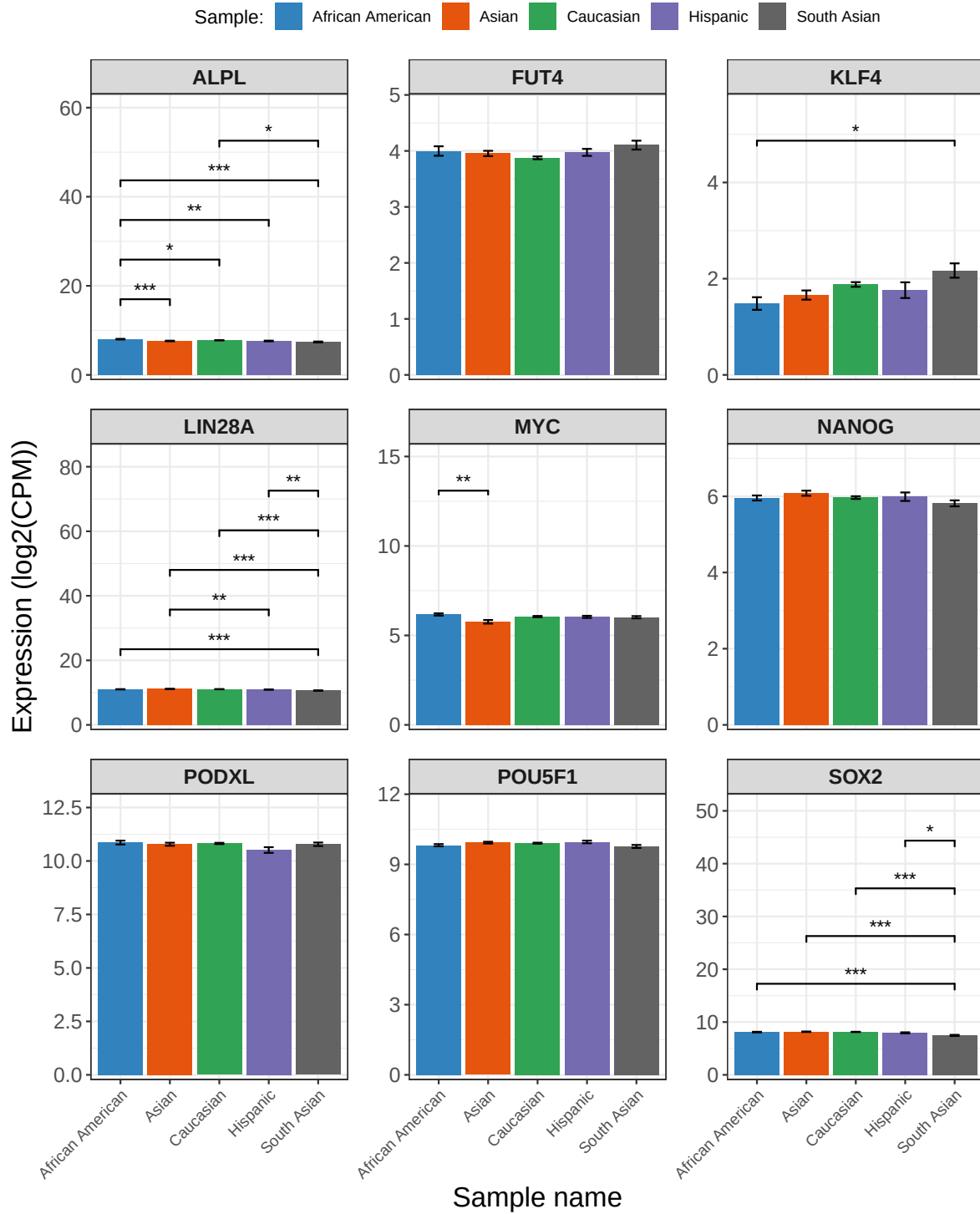


Figure 6: Expression of Pluripotency Genes by Race. Bar plots of log2 count per million gene expression by pluripotency gene. Bars represent mean expression $\pm$ standard error. Pairwise statistical significance was determined using Games-Howell post hoc tests following Welch's ANOVA, and significance was indicated with asterisks: $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***)¹⁰

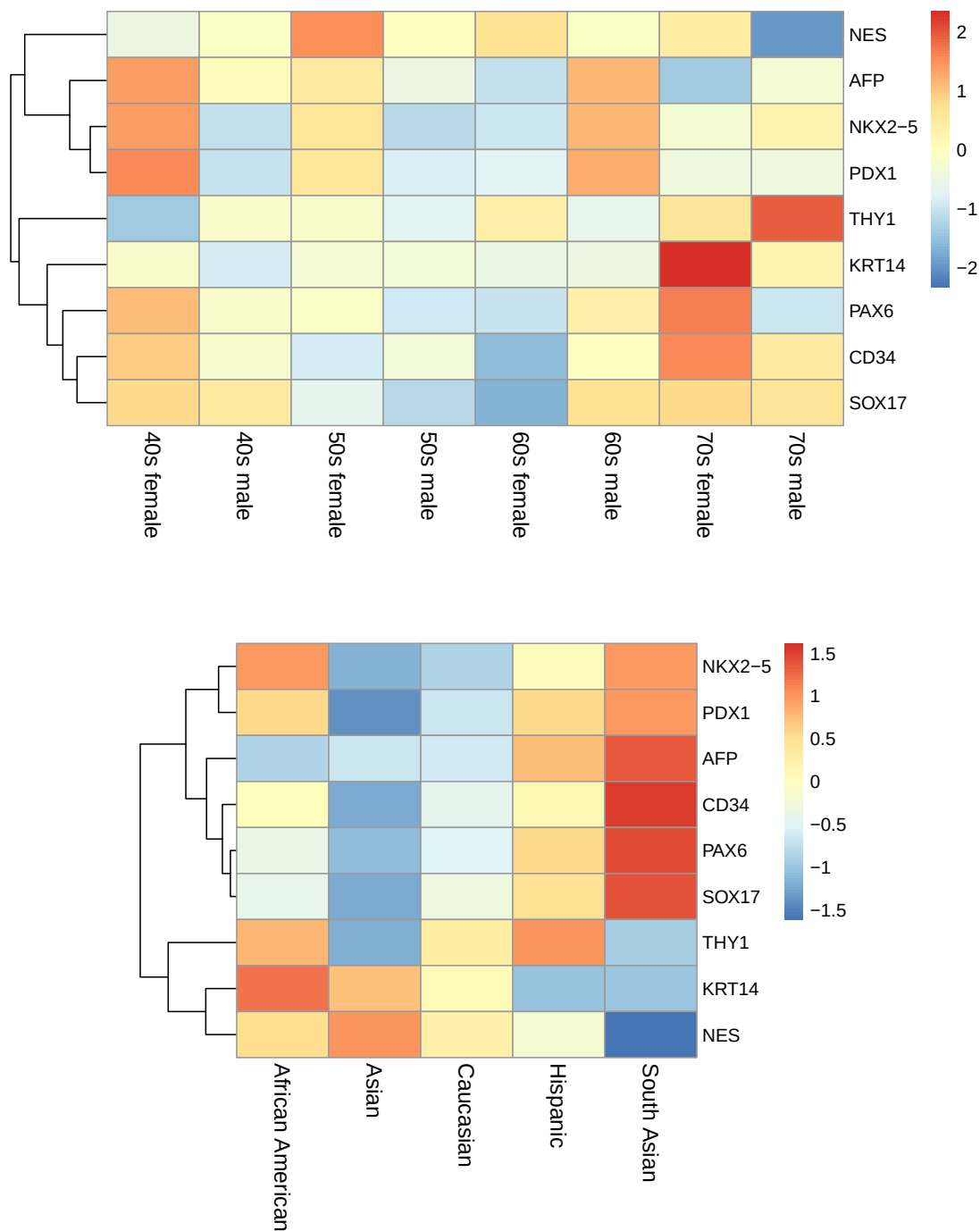


Figure 7: Heatmap of Differentiation Gene Expression. Rows correspond to genes and columns correspond to samples grouped by age-sex (top) and race (bottom). Colors indicate relative expression levels based on log2 count per million expression. Hierarchical clustering was applied to the genes to highlight patterns of expression.

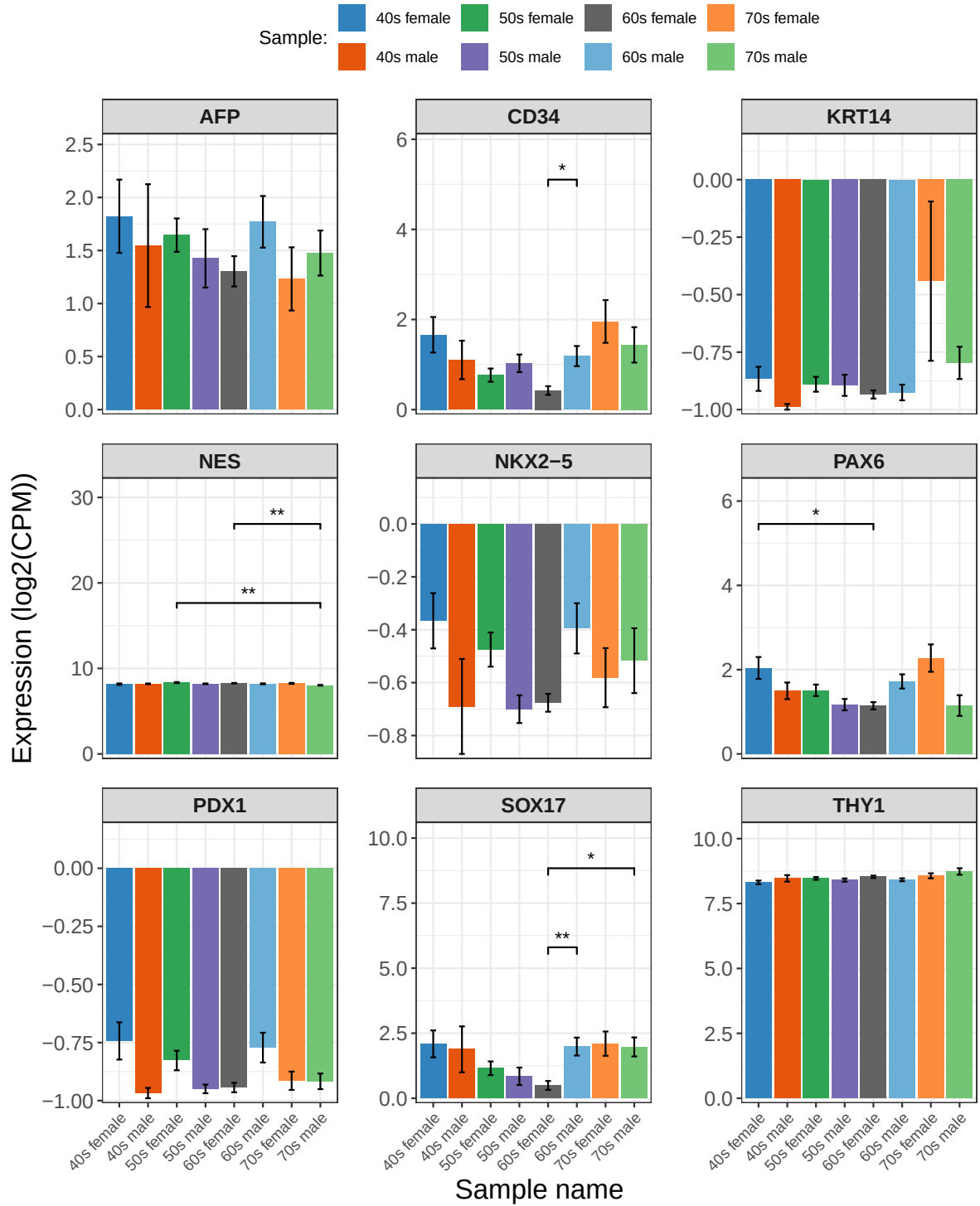


Figure 8: Expression of Differentiation Genes by Age-Sex. Bar plots display log2 count per million gene expression by differentiation gene. Bars represent mean expression $\pm$ standard error. Pairwise statistical significance was determined using Games-Howell post hoc tests following Welch's ANOVA, and significance was indicated with asterisks: $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***) .

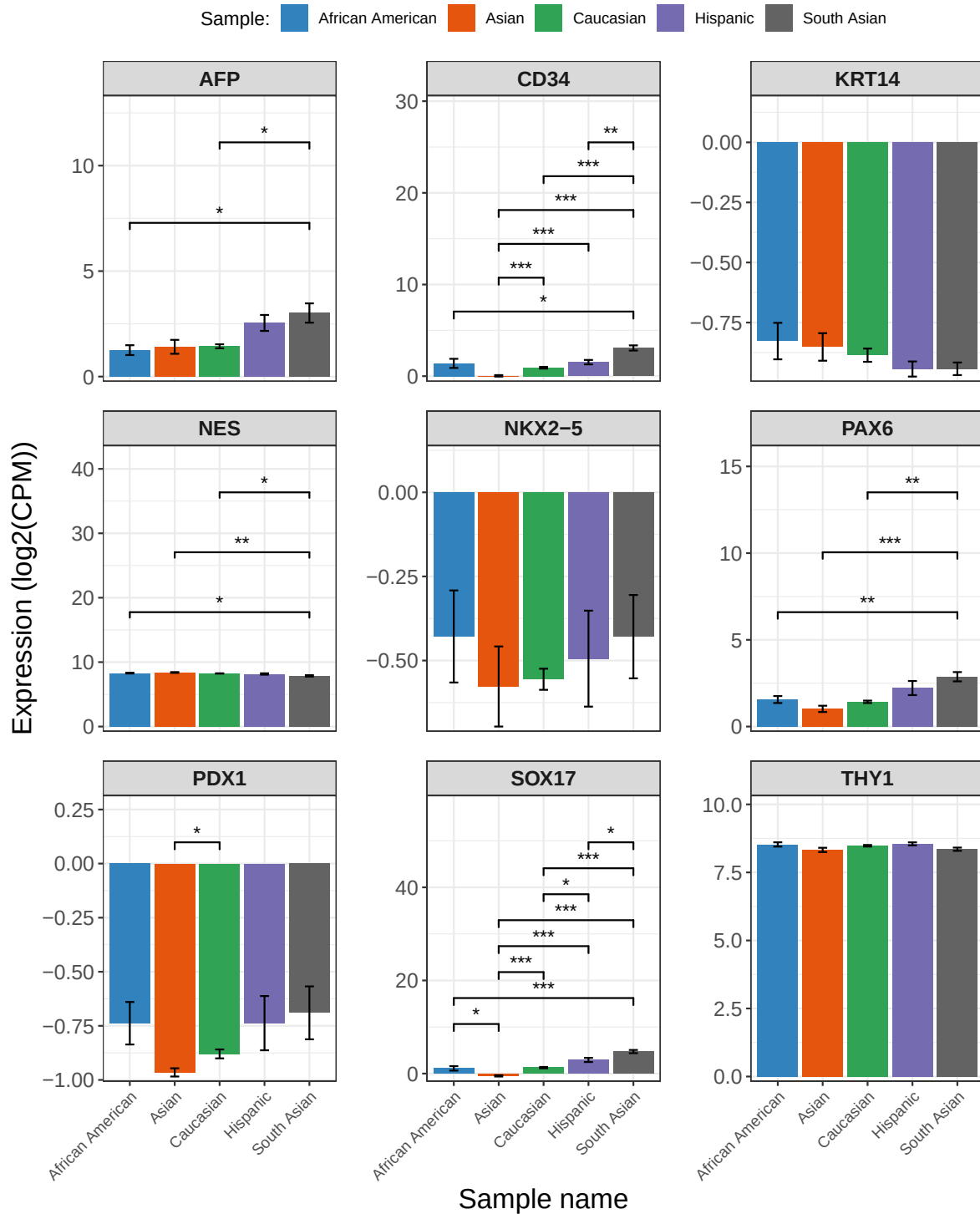


Figure 9: Expression of Differentiation Genes by Race. Bar plots display log2 count per million gene expression by differentiation gene. Bars represent mean expression $\pm$ standard error. Pairwise statistical significance was determined using Games-Howell post hoc tests following Welch's ANOVA, and significance was indicated with asterisks: $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***).

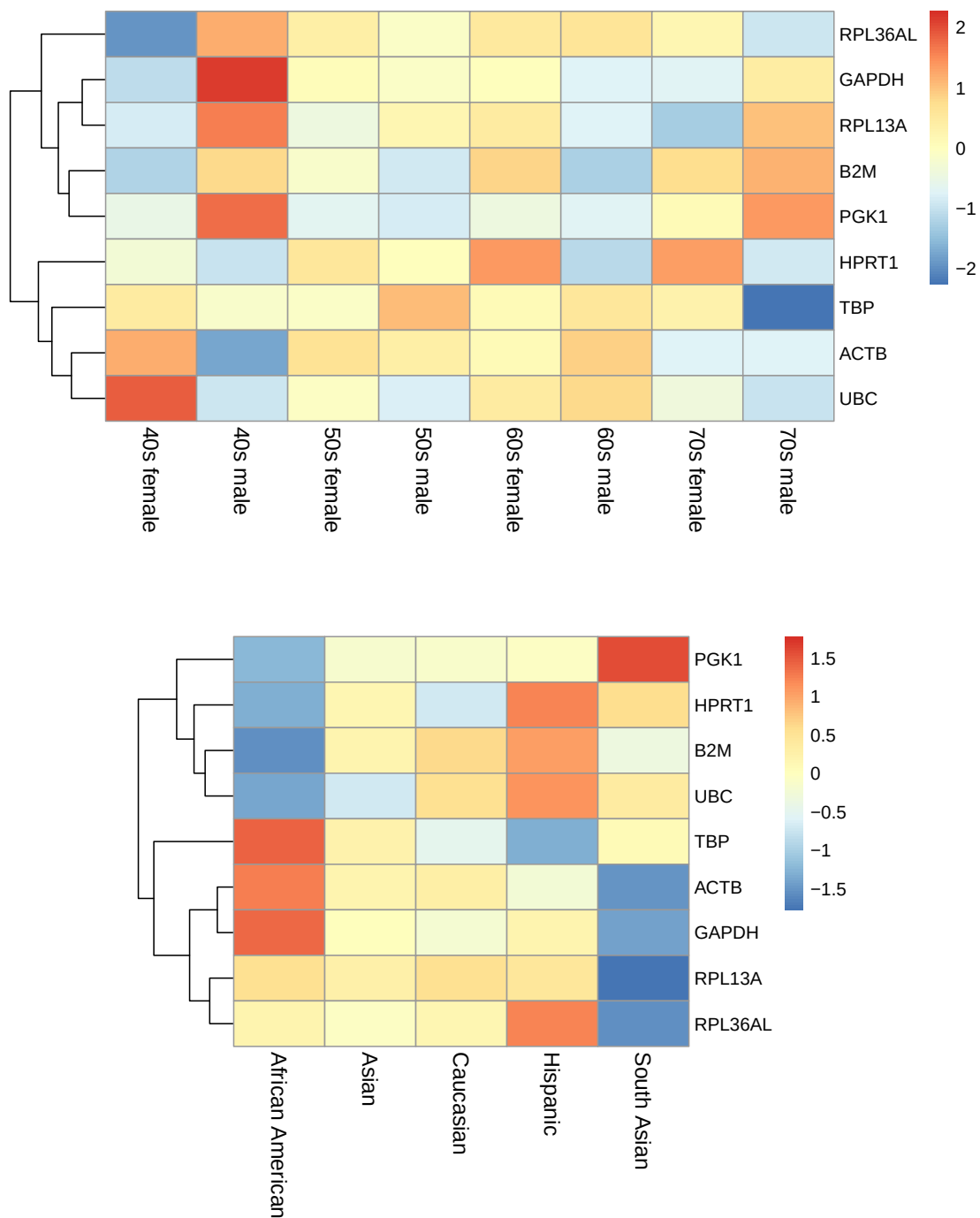


Figure 10: Heatmap of Housekeeping Gene Expression. Rows correspond to genes and columns correspond to samples grouped by age-sex (top) and race (bottom). Colors indicate relative expression levels based on log2 count per million expression. Hierarchical clustering was applied to the genes to highlight patterns of expression.

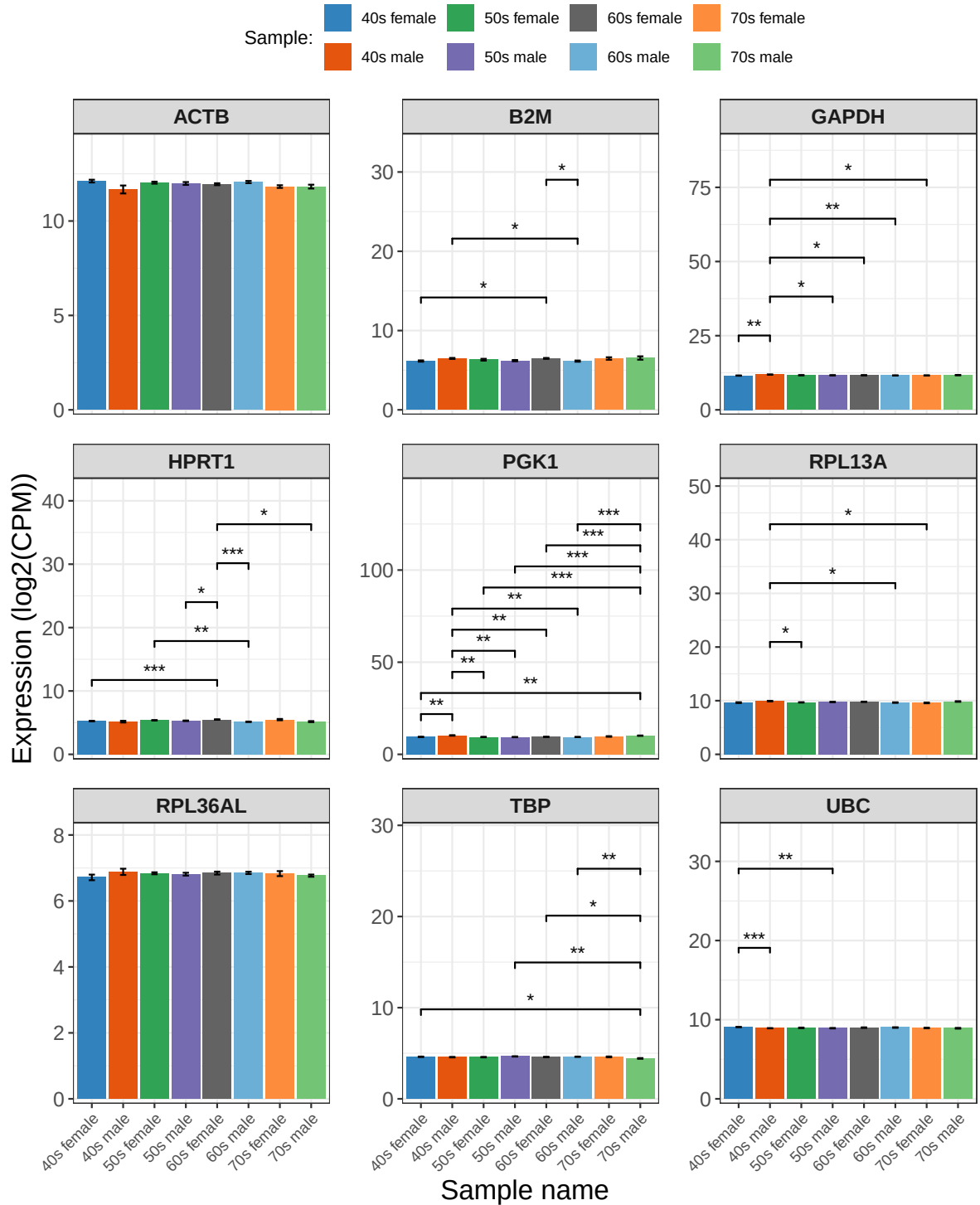


Figure 11: Expression of Housekeeping Genes by Age-Sex. Bar plots display log2 count per million gene expression by housekeeping gene. Bars represent mean expression $\pm$ standard error. Pairwise statistical significance was determined using Games-Howell post hoc tests following Welch's ANOVA, and significance was indicated with asterisks: $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***).

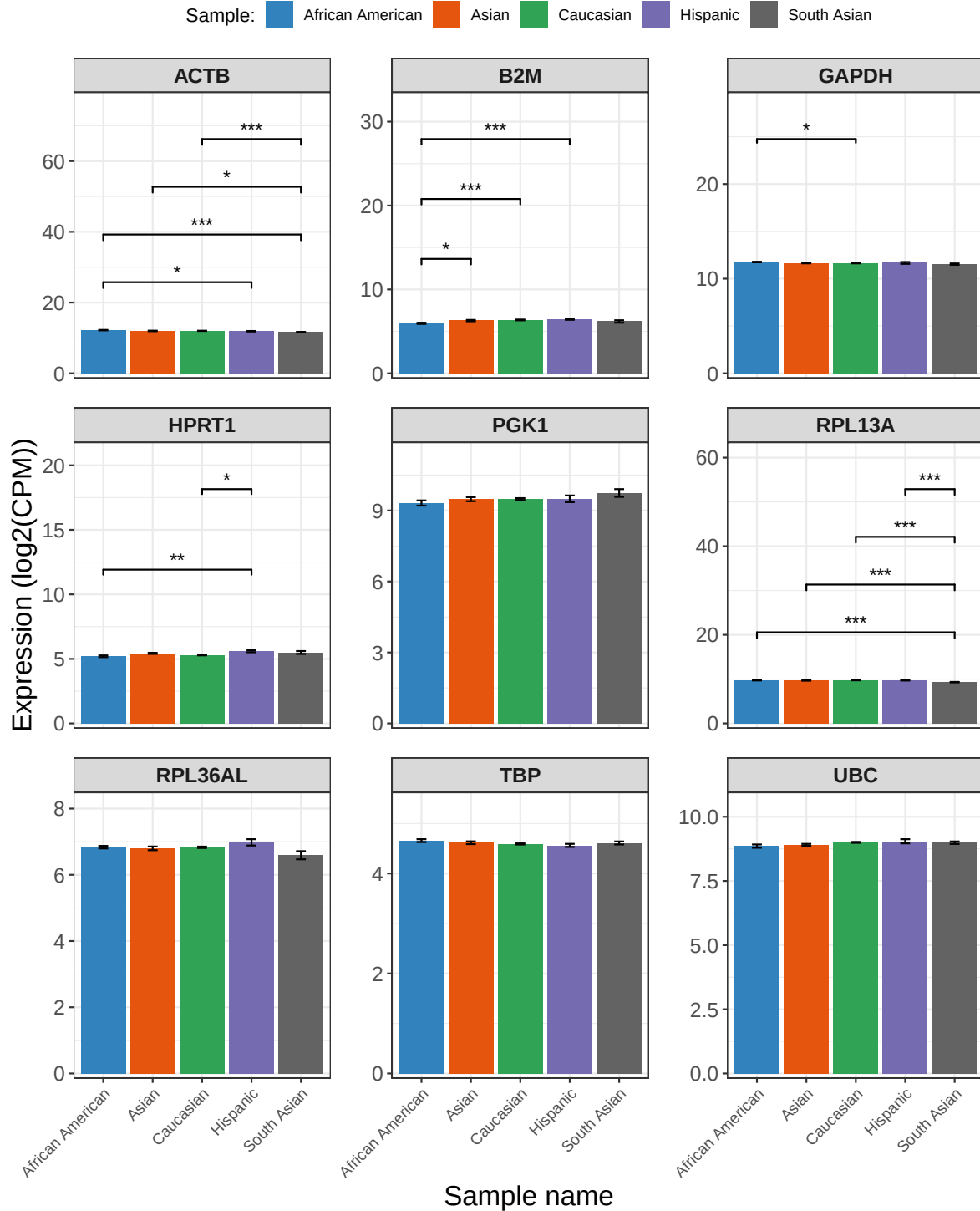


Figure 12: Expression of Housekeeping Genes by Race. Bar plots display log2 count per million gene expression by housekeeping gene. Bars represent mean expression $\pm$ standard error. Pairwise statistical significance was determined using Games-Howell post hoc tests following Welch's ANOVA, and significance was indicated with asterisks: $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***).

Discussion

pie charts

The RNA Seq data is composed of a variety of different types of RNA, as seen in Figure 1. 34.8% of read genes are mRNA protein coding genes, whereas the remaining 65.2% of the detected genes are not protein coding and are primary composed of noncoding RNAs and pseudogenes, which are RNA sequences that look like protein coding genes but are not able to be translated into a protein. 25.6% of the detected genes were pseudogenes. 25.5% were long non coding RNA sequences and 8.5% were small RNAs, which included snoRNA, snRNA, scaRNA, sRNA, and ribozyme RNA. 5.7% were “Other” RNAs, which include ribosomal RNA, immunoglobulin genes, T-cell receptor genes, and mitochondrial RNAs. The right pie chart, on the other hand, displays the expression of these groups. For each gene for each sample, a count per million number was provided by Carcamo-Orive *et al.*. This is typically calculated by dividing the raw count of the gene by the total reads in the sample and multiplying by one million. The total expression was summed for each category, and this was graphed in Figure 2, which displays that 91.7% of the expression is from protein coding genes. The remaining 8.3% accounts for the expression of small RNAs, long non coding RNA, pseudogenes, and “Other” genes. This is supported by _____ data from _____ article, which saw _____.

pca

welch’s anova (one way analysis of means (not assuming equal variance))

PC1 and sex ($p = 0.3892$) PCA2 ($p = 0.6077$)

PCA1 and age_Decade ($p = 0.09443$) and PCA2 ($p = 0.2645$)

PCA1 and age_sex ($p = 0.1993$) and PCA2 ($p < 0.001$)

PCA1 and race ($p < 0.001$) and PCA2 ($p < 0.001$)

When tested independently, neither age nor sex was significantly associated with PC1 or PC2.

However, when individuals were grouped by age–sex categories, a significant effect emerged (Welch’s ANOVA $p =$), suggesting that specific combinations of age and sex capture multi-variate expression patterns not explained by either variable alone.

top 20 genes:

sex linked RPS4Y1 Ribosomal protein S4, Y-linked. Present only in males. High variance indicates sex is a strong driver. XIST X-inactive specific transcript. Expressed in females for X-chromosome inactivation. Sex-linked. DDX3Y DEAD-box helicase, Y-linked. Male-specific. TSIX Antisense of XIST, regulates X-chromosome inactivation. Female-specific. TXLNGY Y-linked, involved in transcription/translation regulation. KDM5D Lysine demethylase 5D, Y-linked. Male-specific. USP9Y Ubiquitin-specific peptidase 9, Y-linked. Male-specific. EIF1AY Eukaryotic translation initiation factor 1A, Y-linked. Male-specific. UTY Ubiquitously transcribed tetratricopeptide repeat, Y-linked. Male-specific. NLGN4Y Neuroligin 4, Y-linked. Male-specific. RPS4XP22 Pseudogene similar to RPS4X/RPS4Y.

other genes CST1 Cystatin SN, extracellular protease inhibitor. May vary in differentiation. CER1 Cerberus, secreted factor involved in early development and differentiation. LARGE1 Glycosyltransferase, involved in post-translational modification. NLRP2 NLR family pyrin domain containing 2, involved in early development, oocyte/embryo regulation. COL3A1 Collagen type III alpha 1, ECM protein, differentiation-related. ZNF208 Zinc finger protein, transcription factor, potential regulatory role. APOA1 Apolipoprotein A1, lipid transport, may reflect metabolic differences. MEG3 Long non-coding RNA, tumor suppressor, epigenetic regulation, differentiation-related. LINC00261 Long non-coding RNA, involved in endoderm differentiation.

Pluripotency, Differentiation, and Houskeeping Gene Expression

The expression of pluripotency genes, SOX2, OCT4 (POU5F1), KLF4, c-Myc (MYC), TRA-1-81 and 1-60 (PODXL), NANOG, SSEA4 (FUT4), LIN28A, and Alkaline Phosphatase (ALPL), were analyzed after grouping by both age-sex and race. These genes include transcription factors, SOX2, OCT4, KLF4, and NANOG, involved in maintaining a pluripotent state, cell surface markers, SSEA4 and TRA-1-81/TRA-1-60, RNA binding protein, LIN28A, and a highly expressed enzyme in pluripotent cells, Alkaline Phosphatase. These genes have been found to be highly expressed in pluripotent cells, such as iPSCs., (_____source).

To determine if the expression of these pluripotent genes varied across conditions, two linear mixed models were used. Expression was modeled as a function of age-sex, with random intercepts grouped by gene to account for baseline differences in gene expression, where j represents the observation of each gene, i , and u represents the part of the intercept that is unique to each gene.

$$\widehat{Expression}_{ij} = \beta_0 + \beta_1(age_sex_{ij}) + u_{0j} + \varepsilon_{ij}$$

The same model was then run, but using race as the fixed effect. In both cases, Type III Analysis of Variance Table with Satterthwaite's method were used to determine which groups

had statically significance differences in gene expression when compared to the reference groups (female 40s and African American).

Residual variance was 0.189 and 0.190 when grouping by age-sex and race, respectively, and gene intercept variance was 9.82 for both. This suggests that the majority of the variance is between genes, not between groups. It was found that 70s female group displayed a 6.5% decrease in pluripotency expression (-0.097 , $p < 0.05$.), whereas 70s male saw a 13% increase in pluripotency expression ($+0.179$, $p < 0.001$).

no difference between other groups and the reference group (40s female)

Type III Analysis of Variance Table with Satterthwaite's method - confirms age_sex have a significant effect on pluripotency gene expression when averaged across all genes. $p < 0.001$.

FUT4 showed 11 significant pairwise comparisons across age-sex groups after FDR correction (Games-Howell test), LIN28A showed 9, and NANOG, POU5F1, and SOX2 with 5 significant pairwise comparisons.

KLF4 and ALPL showed the fewest significant differences, with only one pairwise comparison reaching significance.

all gene has significant pairwise comparisons

linear mixed model Expression $\sim$ race + 1|Gene name

Residual variance = 0.1901 and gene intercept variance = 9.82 means most of the variation is between genes, not race. the genes differ a lot. residual noise is small, model fits wells

South Asian $2^{(-0.143)} = 0.9 = 10\%$ decrease in expression. ($p < 0.01$)

After adjusting for gene-level random effects, South Asian individuals show significantly lower mean log gene expression ($= -0.143$, $p = 0.0059$).

This corresponds to approximately a 10% decrease in pluripotency gene expression relative to the reference race group.

ALPL and LIN28A showed 5 significant pairwise comparisons, and SOX2 with four.

Several pluripotency genes, including FUT4, NANOG, PODXL, and POU5F1, did not show any significant pairwise differences across age-sex groups, indicating relatively stable expression.

Differentiation Markers

Expression of Differentiation Genes (ectoderm: PAX6, NESTIN (NES) (neural progenitor/stem cell markers), KRT14 (epithelial), endoderm: SOX17, AFP (hepatic), PDX1 (pancreatic), mesoderm: CD34 (hematopoietic), CD90 (THY1)(mesenchymal), NKX2.5 (cardiac progenitor)).

age_sex

random effects:

Gene-level intercept variance = 13.168 (SD = 3.629), showing substantial variation between genes.

Residual variance = 1.195 (SD = 1.093), within-gene variation.

fixed effects

50s female, -0.269, $p < 0.001$ 17% decrease

50s male -0.398, $p < 0.001$ 24.5% decrease

60s female -0.500, $p < 0.001$ 29.3% decrease

Compared with 40s female samples, differentiation gene expression was 17% lower in 50s female, 24.5% lower in 50s male, and 29.3% lower in 60s female samples ($p < 0.001$ for all). No significant differences were observed for 40s male, 60s male, 70s female, or 70s male.”

race

Intercept 13.169 Residual 1.163

Asian, -0.42975, 25.3% decrease $p < 0.001$

Hispanic, + 0.39902, 32.0% increase, $p < 0.01$

South Asian, + 0.85435, 80.0% increase, $p < 0.001$

Housekeeping Genes

Figure 5. Expression of Housekeeping Genes (some sort of general figure or sorted by house-keeping type (i.e. ribosomal, metabolic, cytoskeleton, etc.))

age_sex

Intercept 7.163 Residual 0.150

40s male, + 0.1325, 9.6% increase, $p < 0.001$

50s female, + 0.03375, 2.4% increase, $p < 0.01$

60s female, + 0.08191, 5.7% increase, $p < 0.01$

race

Intercept 7.1635 Residual 0.1507

Hispanic, + 0.09717, 7.1% increase, $p < 0.05$

Conclusion

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